SEA-VQA: Visual Question Answering for Kenyan Cultures Across Swahili, English, and Ateso

Ashley Meier[†], Olatunbosun Lapite[†], Amakalu Vitalis[‡]
[†]Princeton University, [‡]Maseno University

Introduction

Vision-Language Models (VLMs) have shown strong performance on **Visual Question Answering** (VQA) in languages such as English, Spanish, and Chinese, but their ability to reason about culturally grounded visual content in **under-represented African languages** remains underexplored. This project aims to help close the digital gap left behind by auditing and comparing VQA models.

- **Motivation:** Kenyan cultural knowledge is limited in web-scale datasets, and languages such as **Ateso** remain under-represented in multilingual vision-language evaluation.
- **Gap:** Existing VQA benchmarks such as **Afri-MCQA and CVQA** provide limited coverage of locally grounded cultural images across **Swahili, English, and Ateso**.
- **Contribution:** We evaluate **six VQA models** on a locally curated dataset of Kenyan cultural images and questions in **Swahili, English, and Ateso**, testing how well current VLMs transfer across languages with different levels of digital representation while reasoning about culturally specific visual content.

Data Collection and Captioning



- Over **1,000 culturally relevant images** were captured across Kenya to maximize representation across CVQA categories. A portion of these images were captioned multilingually with detailed descriptions of their **location, visual content, and cultural significance.** These captions were then utilized to support question generation.

VQA Creation

Exemplar Image

1. What attire is being worn in this image?
   a. **Owali sisal skirts**
   b. Hula kahiko skirts
   c. Piupiu skirts
   d. Liku

Question #1

2. What community is this attire representative of?
   a. **Luo**
   b. Maasai
   c. Somali
   d. Meru

Question #2

3. When and for what is this attire traditionally worn?
   a. **By women and used for dancing. It is primarily worn during marriage ceremonies, celebrating newborns, and initiations. The loose fibers allow for free movement and for the skirt to sway and flow to the music.**
   b. By children and used for dancing. It is primarily worn at children's birthday parties and celebrates the freedom of youth and celebration of life. Children dance in circles and play the nyatiti and abu
   c. By women and men and used for rituals. It is primarily worn during sacred moments for religion and coming of age. The community believes its material represents the creations of the Divine, and is thus holy attire.
   d. By women and used for fashion. It is primarily worn to show off the vibrant colors visible, such as bright pink, purple, and teal. The stripping, softening, and dyeing of fibers from the sisal plant make it very beautiful, and this fine craftsmanship makes it an attire worn for fashionable and flashy purposes.

Question #3

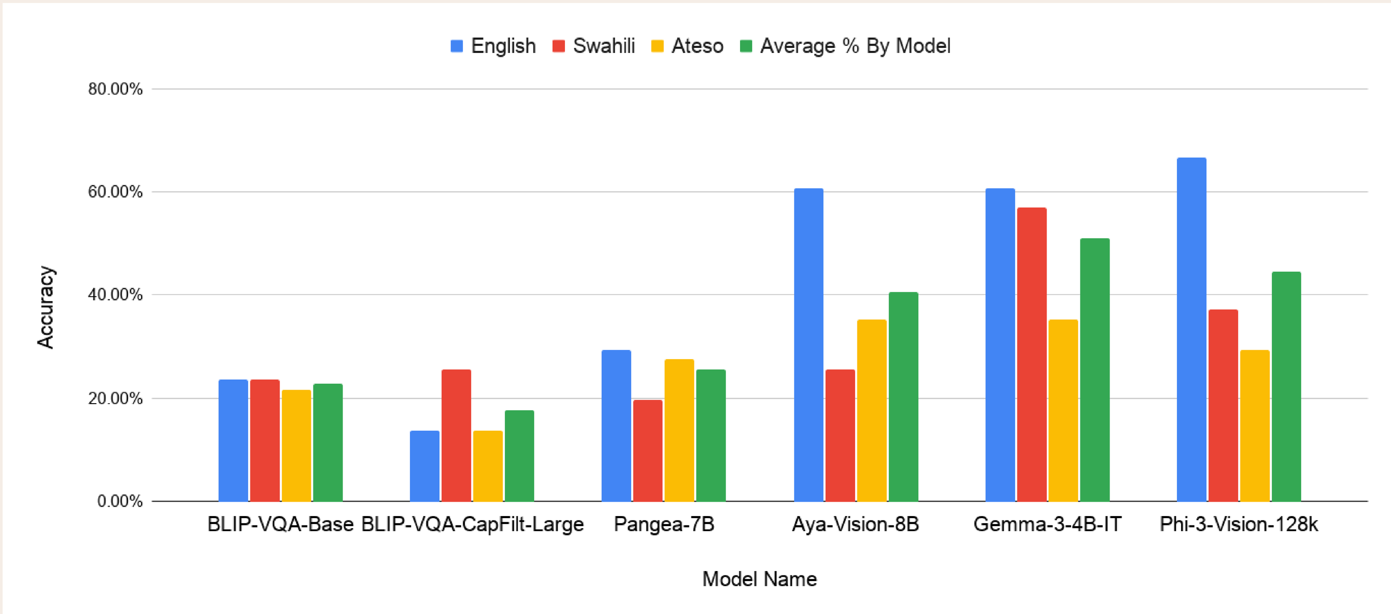
Exemplar Image Questions

- **51 culturally relevant questions** were developed from a curated set of **14 images** (selected from the larger dataset) and translated into **Swahili, English, and Ateso.** The image captions were used to identify key visual details and determine the correct answer, while culturally and visually similar objects informed the distractor (incorrect) multiple-choice options.

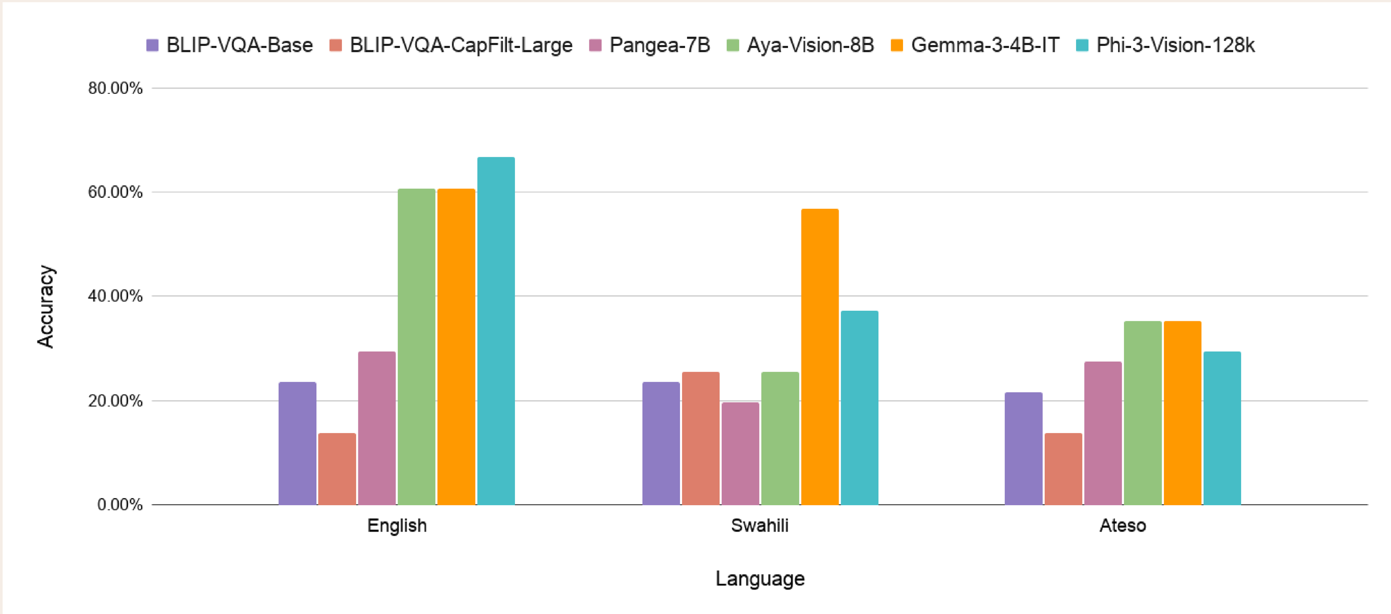
Model Configurations

- The six models, **BLIP-VQA-Base, BLIP-VQA-CapFilt-Large, Gemma-3-4B-IT, Pangea-7B, Phi-3-Vision-128k, and Aya-Vision-8B,** were utilized. These models were selected due to their multilingual capabilities, visual question answering specialization, and multicultural training sets.

Results



Performance By Model



Performance By Language

Conclusion, Discussion, and Next Steps

- **Conclusion**: Overall, the evaluated models demonstrated **limited ability** to answer visual questions about Kenyan culture. For the two BLIP models and Pangea, given that each question had four answer choices **(25% chance accuracy)**, these results indicate performance close to random guessing, suggesting that language was not a primary factor for these models. Aya, Gemma, and Phi performed comparatively better, with average accuracies between **40–50%.** Gemma achieved the highest overall accuracy and the strongest performance in Swahili, while Phi performed best in English. Despite these differences, only Gemma exceeded **50% average accuracy,** indicating that current vision-language models struggle with culturally relevant visual question answering, particularly for under-resourced African languages.
- **Next Steps:**
  - Evaluate additional **closed-source and cloud-based** vision-language models.
  - **Increase the dataset size**, conduct more **repeated trials**, and **expand question types** to improve the reliability and generalizability of the results. Test each model more thoroughly, allowing us to average out the scores of each model, removing bias.
- **Limitations:**
  - **Limited time and computational resources** restricted the number of models evaluated and the number of repeated trials.
  - The dataset size **limited the scope** of the analysis.
  - We ran each model through our laptops' GPUs, causing each model to take upwards of **30 minutes** to complete all the questions.